1- A network admin notices that some frames are not reaching their destination on a LAN. No cables are broken and the devices are powered on. The admin captures traffic and sees that certain frames arrive at the NIC but are immediately discarded.

QUESTION

Which Layer 2 mechanism is responsible for this behavior, and what does it check?

2- A warehouse uses old Ethernet hubs connecting barcode scanners. Workers complain that data entry slows down dramatically when multiple scanners fire simultaneously. A technician proposes replacing hubs with switches.

QUESTION

Explain why collisions occur with hubs and how replacing them with switches eliminates this problem.

3- A student examining a data link frame with Wireshark notices a 4-byte field at the very end of every Ethernet frame. She asks what it does and whether it guarantees reliable delivery.

QUESTION

Name this field, describe its function, and explain its limitation.

4- A network monitoring tool flags two types of unusual Ethernet frames on a campus network: some are 48 bytes long and others are 1520 bytes long.

QUESTION

What are these frames called, what is the valid Ethernet frame size range, and what does the switch do with them?

5- A brand-new Cisco switch is just powered on and connected to four PCs. PC-A sends a frame addressed to PC-D. Describe exactly what the switch does at this moment.

QUESTION

Walk through the switch's learn-and-forward process for this first frame.

6- An IT manager is upgrading a hospital's internal VoIP system. A vendor recommends using store-and-forward switches rather than cut-through switches for this deployment.

QUESTION

Why is store-and-forward preferred for VoIP and QoS-sensitive traffic?

7- A junior technician asks why IP is described as "connectionless" and "best effort." He is confused because data seems to arrive reliably when he browses websites.

QUESTION

Explain what connectionless and best-effort mean at the IP layer, and what actually ensures reliable delivery.

8- A user's PC (192.168.1.50/24) can print to a printer on the same subnet but cannot reach any device outside the office. A check reveals the PC has no default gateway configured.

QUESTION

Why can the PC reach local devices but not remote ones, and what is the solution?

9- A network engineer notices a burst of broadcast traffic consuming 30% of LAN bandwidth every 5 minutes. Closer inspection shows these are ARP requests originating from many hosts simultaneously.

QUESTION

Explain why ARP uses broadcasts, what happens when excessive ARP traffic appears, and one security concern.

10- A student setting up an IPv6-only lab segment is puzzled — there is no ARP traffic visible in Wireshark, yet devices successfully communicate. She sees ICMPv6 messages she doesn't recognize.

QUESTION

What replaces ARP in IPv6, and what message types does it use for address resolution?

11- A company deploys both a wired Ethernet network (using a legacy hub) and a Wi-Fi network. During a busy period, the wired segment shows frequent retransmissions; the wireless segment shows high latency before transmissions even begin.

QUESTION

Compare the collision handling mechanisms of each segment and explain why their performance problems differ.

12- A router receives an Ethernet frame containing an IPv4 packet on one interface. It must re-encapsulate the packet and send it over a Point-to-Point (PPP) WAN link. A student asks: does the LLC sublayer change between the two hops?

QUESTION

Trace what happens to the Layer 2 encapsulation at the router and explain the role of the LLC field in identifying upper-layer protocols.

13- Two switches are connected together without Spanning Tree Protocol (STP). Two PCs — PC-A on SW1 and PC-B on SW2 — are communicating. A technician unplugs and reconnects a cable, temporarily creating a loop. The network becomes unresponsive within seconds.

QUESTION

Explain step-by-step how MAC address table instability and broadcast storms arise from this loop.

14- A server connected to a Cisco switch runs at 10/100 Mbps. The switch port is set to full-duplex; the server NIC defaults to half-duplex via autonegotiation failure. Throughput is terrible but no physical errors appear.

QUESTION

Describe what duplex mismatch causes at the frame level, and what symptoms would appear in the switch port counters.

15- An application sends large data packets across a network that includes an IPv6-only segment. A network engineer notices that unlike on IPv4 paths, no routers along the IPv6 path fragment the packets. A large packet that exceeds an intermediate link's MTU is simply dropped.

QUESTION

Explain how fragmentation differs between IPv4 and IPv6, and what mechanism is used instead in IPv6.

16- A router's routing table has three entries: 10.0.0.0/8 via R1, 10.1.0.0/16 via R2, and 10.1.1.0/24 via R3. A packet arrives with destination 10.1.1.45.

QUESTION

Which route is selected and why? What happens if no match exists at all?

17- A company uses static routes to connect a remote branch office to headquarters. One day, the WAN link to the branch fails and traffic is rerouted through a backup ISP link. The routing tables of all routers still show the original static route as active.

QUESTION

Why did the static route not update, and what would a dynamic routing protocol have done differently?

18- PC-A (192.168.1.10/24) needs to send data to Server-B (10.0.0.50). PC-A sends an ARP request for 10.0.0.50 but gets no reply. A network engineer says this is expected behavior.

QUESTION

Why does ARP fail for remote destinations, and what address does PC-A actually use in its ARP request when sending to a remote host?

19- In a large IPv6 enterprise network with 500 devices per segment, a network architect argues that replacing IPv4's ARP with IPv6's Neighbor Discovery significantly reduces unnecessary CPU interrupts on end hosts.

QUESTION

Explain the technical reason why ICMPv6 NS messages are more efficient than ARP broadcasts, referencing the specific address type used.

20- PC-A (192.168.1.10/24, MAC AA:AA:AA:AA:AA:AA) pings Server-B (192.168.2.20/24, MAC BB:BB:BB:BB:BB:BB) through Router-R (interfaces: 192.168.1.1 / 192.168.2.1, MACs CC:CC on LAN1, DD:DD on LAN2).

QUESTION

For the ICMP packet traveling from PC-A to Server-B, describe the Layer 2 (MAC) and Layer 3 (IP) source/destination addresses at each hop: (1) PC-A to Router, (2) Router to Server-B.